Yeqin Liu

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I am a postdoctoral assistant professor at the University of Michigan, working in algebra and algebraic geometry.

Specifically, I am interested in: derived categories of projective varieties and quivers; DG and A_∞ algebras; Bridgeland stability; DM stacks, twisted sheaves and Brauer groups; stable sheaves and moduli spaces; Brill–Noether theory; classical low-dimensional algebraic geometry.

Employment & Education

University of Michigan, Postdoctoral Assistant Professor — Sep. 2024–present
Mentor: Alexander Perry

University of Illinois Chicago, Ph.D. Mathematics — 2019–2024
Advisor: Izzet Coskun

Nanjing University, B.S. Mathematics — 2015–2019

Research

1. *Phantom categories on 3-vertex directed DG quivers*
   arXiv:2609.27191; with Yu Shen
2. *Cartan determinants in algebraic geometry*
   arXiv:2609.29598; with Yu Shen and Ziyu Zhang
3. *A counterexample to a global-dimension bound for weighted projective lines*
   arXiv:2608.23981; with Bochao Kong and Yu Shen
4. *A counterexample to Han's conjecture*
   arXiv:2608.00177, submitted; with Bochao Kong and Yu Shen
5. *A counterexample to DG version of Han's conjecture*
   arXiv:2512.12460, submitted; with Yu Shen
6. *Picard group action on the category of twisted sheaves*
   J. Noncommut. Geom. (2026), published online first; with Ting Gong and Yu Shen
7. *A looming of phantoms*
   Adv. Math. 499 (2026), 111046; with Kimoi Kemboi, Daniel Krashen, Tianle Liu, Eoin Mackall, Svetlana Makarova, Alexander Perry, Antonios-Alexandros Robotis, and Sridhar Venkatesh
8. *Geometric phantom categories do not admit Noetherian t-structures*
   arXiv:2503.02052, submitted
9. *Stringy Chow rings and weighted blow-ups*
   Int. Math. Res. Not. 2026, no. 3, rnaf374; with Qiangru Kuang, Rachel Webb, and Weihong Xu
10. *Morita theory on root gerbes*
    arXiv:2409.20317, submitted; with Yu Shen
11. *Stability conditions on surface root stacks*
    arXiv:2407.19104, submitted; with Yu Shen
12. *Cones of effective cycles on blow ups of projective spaces along rational curves*
    arXiv:2308.13964, submitted; with Benjamin Gould
13. *Locally-freeness of simple rigid sheaves on nodal K3 surfaces*
    arXiv:2307.00091, submitted
14. *Nonexistence of exceptional bundles on $\mathbb{P}^3$ with maximal possible ranks*
    J. Algebra 695 (2026), 283–310
15. *The cohomology of spherical vector bundles on K3 surfaces*
    arXiv:2210.11030; preprint being polished
16. *Higher rank Brill–Noether theory on $\mathbb{P}^2$*
    Int. Math. Res. Not. 2023, no. 24, 22096–22137; with Benjamin Gould and Woohyung Lee

Awards

UIC Dean's Scholar Fellowship — Fall 2023

Mentoring

Directed Reading Program on quiver representations Spring 2026
Mentees: Yotam Chopra and Xiangnong Wu

Directed Reading Program on algebraic geometry Spring 2020
Mentee: Neelima Borade

Teaching

University of Michigan 2024–2026
Calculus II (Fall 2024), Linear Algebra (Spring 2025), Calculus I (Fall 2025), Multivariable Calculus (Spring 2026), Calculus II (Fall 2026)

University of Illinois Chicago 2019–2023
Mathematical Reasoning (Fall 2019), Elementary Linear Algebra (Spring 2020), Calculus I (Fall 2020), Calculus I (Spring 2021), Calculus I (Fall 2021), Calculus II (Spring 2022), Precalculus (Summer 2022), Mathematical Reasoning (Fall 2022), Calculus I (Spring 2023), Calculus II (Summer 2023), Calculus II (Fall 2023)

Selected Talks

Lectures on Han's conjecture Fall 2026
4-lecture series, Seminario de Álgebra Homológica y Hierbas Afines (SAHHAS), Universidad de Buenos Aires

Geometric phantom categories do not admit Noetherian t-structures Summer 2026
Nanjing University algebraic K-theory conference

A counterexample to DG version of Han's conjecture Spring 2026
AMS 2026 Spring Eastern Sectional Meeting, Boston, Massachusetts

Stability conditions on surface root stacks Fall 2024
Caltech/USC joint algebra & geometry seminar, Pasadena, California

Geometric phantom categories do not admit Noetherian t-structures Spring 2025
Brown University algebraic geometry seminar, Providence, Rhode Island

Nonexistence of exceptional bundles on $\mathbb{P}^3$ with maximal possible ranks Fall 2023
Michigan algebraic geometry seminar, Ann Arbor, Michigan

The cohomology of spherical vector bundles on $K3$ surfaces Winter 2022
BUGCAT 2022, Binghamton, New York

Higher rank Brill–Noether theory on $\mathbb{P}^2$ Summer 2022
GTA 2022, Philadelphia, Pennsylvania

Organized Events

Midwest Algebraic Geometry Graduate Student Conference 2022 Summer 2022
Co-organized with Sixuan Lou, Shravan Patankar, Benjamin Tighe, and Junyan Zhao

Derived Category Student Seminar Spring 2022
Co-organized with Benjamin Gould

UIC Graduate Algebraic Geometry Seminar Spring 2022
Co-organized with Kuang Yu Wu

Gromov–Witten Theory Student Seminar Winter 2021

UIC Graduate Algebraic Geometry Seminar Fall 2021
Co-organized with Kuang Yu Wu